CRACEA: GESTURAL PERFORMANCE

Mary Mainsbridge
University of Technology Sydney
Mary.M.Mainsbridge@student.uts.edu.au

ABSTRACT

This paper outlines the concepts and aims behind the musical performance piece, *Cracea*, composed for bass, drums and gestural instrument, the Telechord. The work forms part of practice-based research into the design and application of gestural systems for musical performance. It also provides a platform to assess the influence of gestural control on my own performance practice as a vocalist and live electronic musician and for developing an understanding of the broader themes concerned with gestural interaction in music.

In gestural performance, the absence of a tangible object requires the performer to develop their unique physical language and refine their movement skills to achieve fine-grained musical control. Drawing on research by movement theorists in dance and embodied interaction design, I explore how gestural systems can contribute to enhanced bodily and kinaesthetic awareness for musicians, facilitating self-discovery and the exploration of new movement patterns.

1. INTRODUCTION

Cracea is the second in a series of six works written for electronic percussion, electric bass and movement-controlled instrument, the Telechord. The virtual instrument links the body's proportions and the ratios between the limbs, torso and head to control hybrid physical models and construct four-part harmonies. Projected visuals amplify the performer's movements, depicting the relationships between key joints as strings or wires. The semi-improvised piece has been performed at the 2014 Sydney Fringe Festival and at the 2014 and 2015 Beams Arts Festival in Sydney.

The Telechord's design incorporates whole body interaction and visual cues to guide gestural interaction for the performer and aid audience comprehension. Although the work is a musical piece, it also crosses over into the categories of instrument, composition and software system, reflecting the overlapping functions that often characterise current digital musical systems.

The primary aim of the work is to stimulate awareness of ingrained bodily habits by encouraging movement awareness through exploration and improvisation. Relying on computer vision to detect continuous motion data, the Telechord utilises remote sensing to minimise behavioural constraints on the performer. Designing an open, adaptable system that translates uninhibited human movement into audio processes is a major goal of this artistic and practice-based research.

2. RELATED WORK

The embodied interaction design approach that frames this research emphasises the value of felt experience in informing decision making throughout gestural interface prototyping for performance applications. This approach draws on the theory of embodied cognition, which highlights the primacy of bodily experiences in shaping thought. Marc Leman (2008) extends this theory to the musical sphere, introducing the term embodied music cognition to affirm the centrality of the body in our experience of music.

Much work has been undertaken in the related fields of audio-visual performance and gestural augmentation of vocal performance that complements the emerging acknowledgement of the multimodal and embodied nature of human perception and cognition. Of particular relevance are pieces that incorporate non-tactile systems in digital musical instrument design, sound installations and dance contexts. Myron Krueger's pioneering installation work, VIDEOPLACE, incorporates a vision-based system that tracks hand, finger and whole body motions to allow individuals to manipulate graphic objects with unencumbered gestures (Krueger 1983). A two-dimensional virtual reality world is created using cameras and projectors to enable multiple users, represented as silhouettes, to interact with digital objects, using a rich set of gestures.

The Very Nervous System (VNS) by David Rokeby also provides a responsive environment that transforms the body of the performer or installation visitor into an instrument through a combination of video cameras, image processors, synthesisers and a sound system. The work strives to create an intimate experience within space scaled to the human form to balance the logical detachment of the computer (Rokeby 2010). Similar ideas are explored in the collaborative solo performance work, *Pikapika*, which seeks to reintegrate physicality into technological music and dance performance (Bahn, Hahn & Trueman 2001). The voice assumes a primary role in guiding improvisations and informing the design of composed instruments and mappings between physical actions and sound.

The voice is considered the body's original and most intimate instrument, bearing the emotional expressive imprint of the performer (Emmerson 2007). This individual body signature is further magnified when the

voice intersects with movement in a motion-controlled interface. The mutual influence of movement and the voice has inspired a diverse range of gesturally augmented vocal systems, including Laetita Sonami's Lady's Glove (Bongers 2000); Michel Waisvisz's The Hands (Waisvisz 1985); Donna Hewitt's eMic (Hewitt 2006) and Sidney Fel's Grassp system (Fels, Pritchard & Vatikiotis-Bateson 2009). These works highlight the range of creative applications for extending the natural capacity of their voice through gestures employed during vocal performance and speech.

3. PERFORMANCE OVERVIEW

Cracea marked a transition in my practice from using a gestural controller that tracks upper body motions to manipulate digital audio effects and trigger MIDI note events to performing with an instrument that encourages improvised and spontaneous movements through whole body interaction. In the piece I controlled the pitch and timbre of hybrid physical models according to the velocity and distance between key joints, indicated in Figure 1. The piece explored notions of expansion and contraction of the limbs in relation to the torso to influence harmonic intervals. Fully extended limbs produced large intervals of octaves, while more contained postures produced major and minor thirds, and perfect fourths and fifths.

I began the performance with a solo improvisation featuring a distorted string sound, altering distortion levels with horizontal head movements. The introduction was intended to establish a clear mapping between sound and action for the audience before the entrance of the rhythm section, which grounded the remainder of the piece in forceful, punctuated percussion.

As the piece progressed, the voice acted as an exciter that shaped the overall tone of each sound. Using this input mode, I could combine the two variables of voice and movement to alter the original physical model, adapting my usual playing style as I sought to co-ordinate both elements.

In the absence of tangible feedback, visual feedback and an interaction metaphor were implemented as design strategies to enhance the nuances available through the gestural system. A string instrument metaphor was used to structure the performance experience. Leon Theremin also relied on a string metaphor to describe the design of the theremin, referring to the hand's distance from an electrode in which "the invisible string could be played by changing the hand from 50 to 10 cm or else through moving the wrist and fingers" (Theremin 1999). Adopting this metaphor became an invaluable aid to guide my movements when performing with the Telechord publicly for the first time.

Visual feedback projected onto a scrim in front of the performers depicted joint position data and the amplitude of the audio signal. It also amplified the geometry of the body's evolving postures, represented in the kinesphere, defined by Rudolf Laban (1988) as the space immediately surrounding the body. This abstracted visualisation was designed to display the virtual strings connecting selected body joints and to provide additional feedback aimed at assisting the performer to calibrate their actions spatially. Both the string metaphor and visual feedback were intended to nurture new forms of musicianship that encourage movement awareness in gestural performance.

4. THE TELECHORD

The Telechord integrates real-time physical modelling synthesis and video generation derived from improvised body postures and movement patterns. The instrument enables pitch control and harmonic layering of physical models. As shown in figure 1, four virtual strings are stretched across the apex joints of the skeleton. As the body moves, the length of the virtual resonating wires surrounding the body change the pitch of four separate tones. The shifting relationships between selected joint positions produce harmonies that reflect the architecture of the human body.

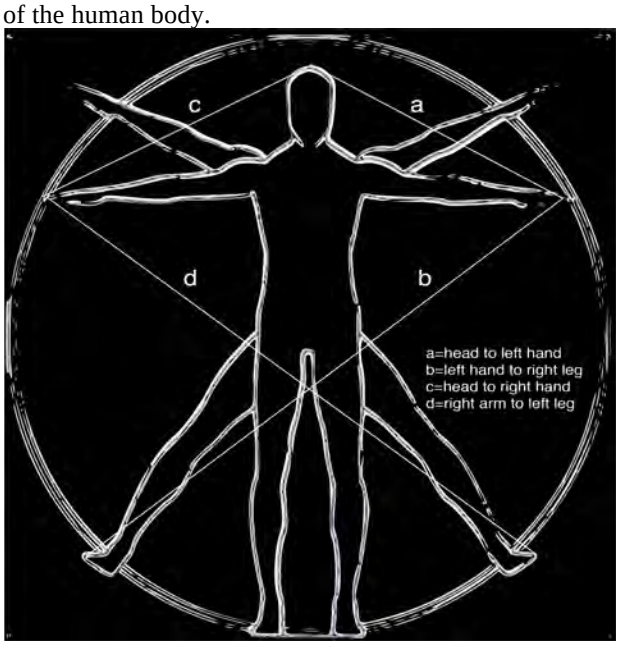


Figure 1. Virtual string positions for the Telechord.

The Telechord can perform several distinct roles in the performances, including functioning as an accompaniment for the voice, augmenting the voice with a range of physical models, and as a solo and ensemble instrument. The system also offers a framework for discovering and experimenting with sound-movement relationships and exploring individual movement tendencies and patterns.

4.1. Audio System

Using IRCAM's Modalys[1] software to simulate a range of vibrating bodies, termed 'objects', a palette of sounds were designed to suit each of the sections in the work. Modalys facilitates virtual instrument design by offering a variety of materials such as plates, strings and

[1] http://forumnet.ircam.fr/product/modalys-en/

membranes, to be matched with exciters including bows and hammers. Using several examples provided with the program as a starting point, alterations were made with LISP before each patch was exported into Max/MSP[1].

The performer has the ability to navigate between separate materials and exciters, morphing between metallic, string and membrane sounds. SimpleKinect[2] middleware is used to send OSC information from the Kinect to the Max/MSP patch transferred from Modalys. As well as detecting position information, the patch also couples acceleration to the excitation of the sound, enabling the performer to gain more nuance from a particular note by varying the speed and intensity of their movements.

The potential of physical models to provide a sonic environment from which to explore gesture has attracted much interest from researchers (Leman et al. 2008). In *Real Sound Synthesis for Interactive Applications* Perry R. Cook writes, "Our evolution and experience in the world has trained us to expect certain sonic responses from certain input gestures and parameters" (Cook 2002). Incorporating these expectations into interaction design potentially offers a familiar grounding drawn from understandings of the physical world.

4.2. Visual Feedback

The visual feedback for the instrument is deliberately simple, comprising lines that visualise the vibrating strings between the performer's joints. These strings change colour in response to the speed and magnitude of the performer movements. Additionally, a particle system displays the amplitude of the instrument's audio output.

The feedback helps to illuminate the mapping for the performer and audience by accentuating the lines and shapes the body forms in relation to the sounds, illustrating its underlying sound-movement mappings.

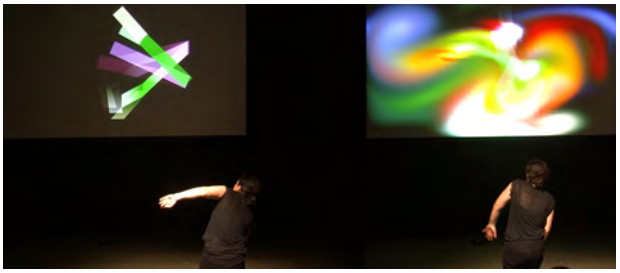


Figure 2. Telechord visualisations.

As the system is always 'on', it is also important for the performer to be able to identify which string is activated so that they can adequately control each separate note of the chord. For this reason the visual feedback is projected at the same scale as the performer's body. The performer sees the projection directly before them through a scrim, yet can still maintain eye contact with the audience.

[1] https://cycling74.com/products/max/
[2] http://deecerecords.com/kinect/

5. KINAESTHETIC AWARENESS

Nurturing kinaesthetic awareness and proprioceptive skills emerged as a core aspect of this practice-based research. Aaron Levisohn (2007) proposes to expand our understanding of the body not only as a controller of multimedia systems but also "as a unique medium unto itself."

Cracea draws on similar approaches to dance-based and virtual reality systems that are created purposely to develop proprioceptive skills, including Thecla Schiphorst and Susan Kozel's work, *whisper: wearable body architectures* (2002), and Levisohn's (2007) mixed reality system that enables users to experience hand movement in a different way by altering visual and aural perception. Like performer/designers who embrace embodied interaction design such as Danielle Wilde (2011) and Jin Moen (2006), I am interested in developing interaction environments where movement can be explored. This design focus encourages a holistic approach to playing a gestural instrument by creating an increased awareness of the movement possibilities available to performers.

The Telechord was initially designed to support a broad range of movement, so that tension is not held in the body through repetitive actions associated with playing traditional acoustic instruments. In musical training, the instrument is often considered superior to the needs of the performer's body. The musician is taught to minimise their movements and become almost transparent to make the instrument 'sing' (McClary 1995). To avoid this type of disembodiment, I deliberately focused on strengthening an awareness of my personal movement patterns. I favoured a design that promoted variations in posture and balanced exertion evenly throughout the body's frame.

The movement explorations encouraged through the design reflect a desire for balance and a sense of self control and body mastery, a valuable skill for musicians wishing to gain proficiency in gestural performance. This process is reminiscent of somatic practices that encourage bodily awareness, balance and conditioning, such as yoga, Feldenkrais and Alexander Technique.

6. REFLECTION

During rehearsals and performances of *Cracea*, I found myself progressively moving beyond my usual movement range and ingrained habits, occasionally crossing into the dance realm. However without formal dance or movement training, I found it challenging to achieve consistent and repeatable sonic results, particularly in the areas of pitch control and harmony. To help calibrate my movements and execute smoother and more accurate pitch transitions, I often visualised the instrument as a physical entity, at times mimicking the expressive circular motions a guitarist makes with their instrument or imagining bowing or plucking the Telechord's virtual strings.

For iconic performers of 'air' instruments like Clara Rockmore, the importance of performing with visual imagery is an essential part of effective technique. Rockmore drew on

a string metaphor to guide performance with the theremin. Similarly, the string metaphor underpinning the Telechord's design helped to structure my performance as I mentally referenced virtual objects to produce more subtle and precise control. Even though the visual feedback illustrated this metaphor through lines connecting active joints, I found myself relying more on an internal mental image of the virtual strings in association with audio feedback to structure my experience.

This technique highlights a two-pronged approach endorsed by somaesthetician, Richard Shusterman (2009), integrating unreflective spontaneous performance with reflective bodily awareness. The problem of using spontaneous movements alone in gestural performance is that they are a product of acquired habits, and without conscious reflection some of these habits can prove damaging. Attending to our movement patterns using body awareness techniques drawn from somatic disciplines and engaging in conscious reflection may contribute to addressing unwanted movement habits. Performing with the Telechord enabled me to engage in a similar process that has improved my sensorimotor skills, awareness of my body during performance, and a willingness to experiment with new movement behaviours to achieve more diverse sonic outcomes.

Perhaps a greater focus on the somatic dimension, as it relates to performance technique and learning, can increase the relevance of new gestural instruments beyond academic and engineering communities. Uniting technical design priorities with attention to somatic aspects that address how the instrument fits the body could thus make gestural systems more relevant to musicians from a broader range of backgrounds (Paine 2015).

7. CONCLUSION

Contributing to existing research on whole body interaction, the piece *Cracea*, written for the Telechord, explored ways to create direct and transparent mappings linking harmonic intervals to the architectural proportions of the body. The string metaphor underpinning the instrument's design informed movement-based control of hybrid physical models activated by voice and movement. Performing the work offered opportunities to explore techniques aimed at encouraging the development of kinaesthetic skills.

First-hand insights throughout the design and rehearsal process also revealed the instrument's influence on expanding movement awareness and range, highlighting the importance of direct engagement with gestural systems in improving movement mastery among musicians with no formal movement training. Through these explorations I developed new ways of moving by firstly observing and then stepping beyond my usual physical patterns to achieve more nuanced expression during movement-based performance.

8. REFERENCES

Bahn, C., Hahn, T. and D. Trueman. 2001. "Physicality and Feedback: A Focus on the Body in the Performance of Electronic Music." *Proceedings of the International Computer Music Conference*, pp. 44-51.

Bongers, B. 2000. "Physical Interfaces in the Electronic Arts." In M.M. Wanderley and M. Battier (eds.) *Trends in Gestural Control of Music*, Paris: Ircam, pp. 41-70.

Cook, P. 2002. *Real Sound Synthesis for Interactive Applications*. MA, US: A K Peters Limited.

Emmerson, S. 2007. *Living Electronic Music*. Aldershot, UK: Ashgate Publishing.

Fels, S., Pritchard, R. and E. Vatikiotis-Bateson. 2009. "Building a Portable, Gesture-to-Audio/Visual Speech System." *Proceedings of the International Conference on Audio-Visual Speech Processing*, pp. 13-18.

Hewitt, D. G. 2006. "Compositions for Voice and Technology." PhD Thesis, University of Western Sydney.

Krueger, M. W. 1983. *Artificial Reality.* MA, US: Addison-Wesley.

Laban, R. 1988. *The Mastery of Movement*. Plymouth, UK: Northcote House.

Leman, M., F. Stynes and N. Bernadini. 2008. "Sound, Sense and Music Mediation: A Historical-philosophical Perspective." In P. Polotti and D. Roccesso (eds.) *Sound to Sense, Sense to Sound: a State of the Art in Sound and Music Computing*, Berlin: Logos Verlag, pp. 15-46.

Levisohn, A. 2007. "The Body as Medium: Reassessing the Role of Kinaesthetic Awareness in Interactive Applications." *Proceedings of Multimedia '07*, pp. 485-488.

McClary, S. 1995. "Music, the Pythagoreans, and the Body." In S. I. Foster (ed), *Choreographing History*. Bloomington: Indiana University Press, pp. 82-105.

Moen, J. 2006. "KinAesthetic Movement Interaction: Designing for the Pleasure of Motion." PhD Thesis, University of Stockholm.

Paine, G. 2015. "Interaction as Material: The Techno-somatic Dimension." *Organised Sound,* 20(1): 82-89.

Rokeby, D. 2010. "Works: Very Nervous System (1986-1990)." Available online at davidrokeby.com/vns.html. Accessed November 2015.

Schiphorst, T. and K. Andersen. "Between Bodies: Using Experience Modeling to Create Gestural Protocols for Physiological Data Transfer." *Proceedings of the CHI 2004 Fringe*, pp. 1-8.

Shusterman, R. 2009. "Body Consciousness and Performance: Somaesthetics East and West." *The Journal of Aesthetics and Art Criticism*, 61(2): 133-145.

Theremin, L. 1999. "Recollections." *Contemporary Music Review,* 18(3): 5-8.

Waiswisz, M. 1985. "The Hands: A Set of Remote MIDI Controllers, *Proceedings of the International Computer Music Conference*, pp. 313-318.

Wilde, D. 2011. "Swing That Thing: The Poetics of Embodied Engagement." PhD Thesis, Monash University.